B&B

BATCH & BUILD · STEM LEARNING SHEET · AGES 8–12

Rock Tumbling: Turning Rough Rocks Smooth

THE SCIENCE

A rock tumbler is a spinning barrel filled with rocks, water, and grit. Grit is a powder made of a super-hard material called silicon carbide. As the barrel turns, the rocks and grit tumble over each other day and night. The grit slowly scrapes away the rough edges, the same way sandpaper smooths wood. You start with coarse grit to shape the rocks, then use finer and finer grit, and finish with polish to make them shine.

Nature does the same thing. Rivers and ocean waves roll stones against sand for years, which is why beach pebbles are smooth and round. This wearing-away is called erosion.

Hardness matters too. Scientists rate minerals on the Mohs scale from 1 (talc, softest) to 10 (diamond, hardest). Rocks of similar hardness should tumble together so the soft ones don't get ground away.

KEY WORDS

Abrasion: wearing something down by rubbing.

Grit: hard powder that scrapes surfaces smooth.

Erosion: wearing away of rock by water, wind, or ice.

Mineral: a natural solid with a crystal structure.

Mohs scale: a 1 to 10 ranking of mineral hardness.

Polish: the final, finest step that makes rocks shine.

EXPERIMENT – TRACK THE CHANGE

Before tumbling, pick 3 rocks. Weigh each on a kitchen scale and photograph it next to a ruler. After each grit stage, weigh and photograph them again. Make a chart of weight over time.

Think about it: During which stage did they lose the most weight?

SAFETY

An adult sets up the plugged-in tumbler and keeps it somewhere dry. Never pour used grit down a sink, because it clogs pipes; let it settle and throw it in the trash.

CHECK YOUR UNDERSTANDING

1. What does grit do?
2. Why use finer grit at each stage?
3. What's the hardest mineral on the Mohs scale?
4. How does nature tumble rocks?
5. Where should used grit go?

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ANSWER KEY (FOR GROWN-UPS)

1. Scrapes rocks smooth.
2. To make the surface smoother and smoother.
3. Diamond.
4. Rivers and waves roll them against sand.
5. The trash, never the sink.

Name: _____